

Chanul Dandeniya

347-626-6214 | cdandeniya1@gmail.com | [LinkedIn](#) | [Github](#) | cdandeniya.vercel.app

EDUCATION

Stony Brook University

BS Computer Science and Applied Math/Statistics (GPA: 3.68/4.0)

Expected Graduation May 2027

EXPERIENCE

Saiera | *Software Development Intern*

Remote | Jun 2025 – Aug 2025

- Built a course-centric multimodal chatbot using the **GPT-4o API** to answer personalized student questions and generate flashcards and quizzes from textbooks, lecture audio, slides, and class notes.
- Developed a document ingestion pipeline processing 100+ files per day using **Amazon S3**, **CloudConvert API**, **Firebase**, and **MongoDB** to support scalable indexing and retrieval.

Research Lab | *Lead Undergraduate Researcher*

Stony Brook, NY | Jan 2025 – May 2025

- Implemented a real-time human-activity recognition pipeline on low-power edge devices using **TensorFlow Lite** and **OpenCV**, supporting continuous motion classification from live video.
- Led a programming team of 4 members in weekly design meetings using **Jira** to monitor progress and meet deadlines.

Amazon | *Software Engineering Apprentice*

Manhattan, NY | Oct 2021 – Oct 2024

- Contributed to backend services for large-scale billing and payment **APIs**, implementing cross-service integrations on **AWS** to support reliable transaction processing for global advertisers.
- Helped migrate legacy billing workflows to microservice-based architectures (**ECS**, **S3**, **DynamoDB**), shipping performance and resiliency updates that reduced retry incidents by 8% in targeted services.
- Implemented and maintained unit/integration test suites using **JUnit**, **Mockito**, and **Pytest**, improving regression coverage by 15% across multiple release cycles.
- Collaborated with a 12-developer team, performed 40+ code reviews, and contributed to **CI/CD** improvements in **Amazon CodePipeline**, including **Blue/Green deployment** support to reduce release risk.

Amazon Alexa-Speech | *Fellow*

Remote | Jun 2021 – Oct 2021

- Supported development of Alexa speech-processing features by implementing keyword-processing functionality and refactoring **data pipelines** with efficient **data structures**, improving latency/reliability across distributed **microservices** and increasing throughput.

PROJECTS

Chiral Network | *Rust, TypeScript, Svelte, P2P Networking, DHT* | [Website](#) | [Github](#)

- Contributed to a decentralized, BitTorrent-like P2P file-sharing platform with DHT-based peer discovery and multi-source downloads, built with Rust services and a Svelte/TypeScript client.
- Implemented DHT file-access request handling (RequestFileAccess + pending request tracking) and added download chunk integrity verification (tests + runtime checks), improving protocol correctness and reliability.

NovaTrade | *Java, JSP, Servlets, JDBC, MySQL, Tomcat, Maven* | [Website](#) | [Github](#)

- Built a full-stack stock trading web application with **role-based access control** (Manager, Customer Rep, Customer), supporting portfolio tracking, order execution, and sales reporting.
- Implemented advanced order types (**Market**, **Trailing Stop**, **Hidden Stop**) using a layered Controller/Service/Repository architecture with **JDBC DAOs** and **MySQL**, applying prepared statements and input validation for security.

Cruise | *React Native, Expo, TypeScript, Google Maps API, OpenAI GPT-4, iOS* | [Github](#)

- Building a voice-first iOS navigation app with turn-by-turn routing, real-time GPS tracking, and a Google Maps-style UI using React Native (Expo) and the Google Maps/Directions/Places APIs.
- Implemented an AI voice assistant flow with speech-to-text, text-to-speech, and natural-language command handling (e.g., destination search, nearby places, ETA) using OpenAI GPT-4 and Expo Speech/Location services.

TECHNICAL SKILLS

Languages/Frameworks: Java, Python, TypeScript/JavaScript, SQL; React Native), React/Next.js, Node.js

Tools: AWS, MySQL, MongoDB, Firebase, Git Version Control, Jira, JUnit, Mockito, Pytest

Skills: Agile Development, CI/CD, Unit Testing, Microservices, Backend Engineering